James Stainer

I build forecasting systems and data pipelines on large, real-world data, and test how well frontier AI models reason.

London, UK. Open to machine learning, data science and AI evaluation roles.
jamsta.in | stainerjames@hotmail.com | LinkedIn | Google Scholar

Selected work

Forecasting vital signs in paediatric intensive care

PhD thesis, UCL and Great Ormond Street Hospital

I used a PyTorch StemGNN model to forecast heart rate and blood pressure together from 5-second bedside-monitor data, treating the dependencies between them as a learned graph.

I also built the clinical time-series pipeline behind it, which took 4,120 raw admissions down to 364 high-quality patients and about 12.5M observations. It covers age-aware normalisation, missing data, outlier detection and sliding-window segmentation.

Error on 5-minute-ahead forecasts at 1-minute resolution (MAPE, lower is better)

Model	Heart rate	Blood pressure
StemGNN	**2.30%**	**3.49%**
ARIMA	2.82%	3.85%

DrugnomeAI: predicting which human genes can be drugged

With AstraZeneca's Centre for Genomics Research. Published in *Communications Biology*, 2022

An ensemble machine-learning framework that scores all 19,846 human protein-coding genes for druggability, using 324 features from 15 sources. Specialised models are stratified by drug modality (small molecule, antibody, PROTAC) and by therapeutic area.

Top-ranked genes were significantly enriched for targets already in clinical development, and for genome-wide significant hits in a UK Biobank PheWAS of 450K participants. The framework outperformed three published benchmarks.

Paper GitHub Web app

Adversarial evaluation of frontier AI agents

Freelance, current

I write hard bioinformatics and scientific coding tasks that benchmark how frontier LLM agents reason, each with a reference solver, a grading contract and a deliberate reasoning trap. I then audit the agent transcripts for answer leakage, shortcuts and data bugs, so a failure means the model got it wrong and not that the task was broken.

Experience

Dec 2023 to present

Freelance AI Evaluator and Data Science Consultant

Remote

AI evaluation and data science across multiple client projects: model-output review, scientific reasoning benchmarks, failure-mode analysis and applied AI workflows.

Oct 2022 to Apr 2023	**Data Science Intern** AstraZeneca, Centre for Genomics Research, Cambridge Built a PubMed-to-Elasticsearch pipeline indexing over 35M biomedical citations and abstracts. Added NLP pipelines, using BioWordVec embeddings, that ranked candidate drug-target-disease relationships for expert review in place of broad manual search.
Jun 2020 to Sep 2020	**Data Science Intern** AstraZeneca, Centre for Genomics Research, Cambridge Co-developed the DrugnomeAI models, working across the ML pipeline from data collection and feature engineering through to training, validation and benchmarking.

Education

2021 to 2026	**PhD, AI-enabled Healthcare** University College London. Viva passed October 2026 with minor corrections.
2020 to 2021	**MRes, AI-enabled Healthcare** University College London
2019 to 2020	**MSc, Data Science** Lancaster University
2016 to 2019	**BSc (Hons), Mathematics** Lancaster University

Publications and presentations

Raies A, Tulodziecka E, **Stainer J**, et al. DrugnomeAI is an ensemble machine-learning framework for predicting druggability of candidate drug targets. *Communications Biology* 5, 1291 (2022).

Stainer J, et al. Poster, IEEE Engineering in Medicine and Biology Society (EMBS) Conference, 2024.

Stainer J, et al. Poster, IEEE International Conference on Biomedical and Health Informatics (BHI), 2024.

Skills

Languages	Python, SQL, R, Bash
Machine learning	PyTorch, TensorFlow, scikit-learn, graph neural networks, NLP, ensemble methods
Methods	Time-series forecasting, statistical and Bayesian inference, change-point detection
Data and infrastructure	Data pipelines, pandas, NumPy, Elasticsearch, feature engineering, SLURM, Git